

$$\begin{aligned}
 1) \quad a - \int \frac{2-x}{x^2+1} dx &= \int \frac{2}{x^2+1} dx - \int \frac{x}{x^2+1} dx \\
 &= 2 \int \frac{1}{x^2+1} dx - \frac{1}{2} \int \frac{2x}{x^2+1} dx \\
 &= 2 \arctg(x) - \frac{1}{2} \cdot \ln|x^2+1| + C \\
 &= 2 \arctg(x) - \frac{\ln|x^2+1|}{2} + C
 \end{aligned}$$

$$\begin{aligned}
 b - \int (x+1)^2 \ln(x) dx &= \int (x^2+2x+1) \ln(x) dx \\
 &= \int x^2 \ln(x) dx + \int 2x \ln(x) dx + \int \ln(x) dx \\
 &= \frac{x^3}{3} \ln(x) - \frac{x^3}{9} + x^2 \ln(x) - \frac{x^2}{2} + x \ln(x) - x + C
 \end{aligned}$$

$$\begin{aligned}
 L. \int x^2 \ln(x) dx &= \frac{x^3}{3} \ln(x) - \int \frac{x^3}{3} \cdot \frac{1}{x} dx \\
 &= \frac{x^3}{3} \ln(x) - \frac{1}{3} \int x^2 dx \\
 &= \frac{x^3}{3} \ln(x) - \frac{1}{3} \left( \frac{x^3}{3} + C \right) \\
 &= \frac{x^3}{3} \ln(x) - \frac{x^3}{9} + C
 \end{aligned}$$

$$\begin{aligned}
 2. \int 2x \ln(x) dx &= 2 \int x \ln(x) dx = 2 \left( \frac{x^2}{2} \ln(x) - \int \frac{x^2}{2} \frac{1}{x} dx \right) \\
 &= 2 \left( \frac{x^2}{2} \ln(x) - \frac{1}{2} \left( \frac{x^2}{2} + C \right) \right) \\
 &= x^2 \ln(x) - \frac{x^2}{2} + C
 \end{aligned}$$

$$\begin{aligned}
 3. \int \ln(x) dx &= \int 1 \cdot \ln(x) dx \\
 &= x \ln(x) - \int x \frac{1}{x} dx \\
 &= x \ln(x) - \int 1 dx \\
 &= x \ln(x) - x + C
 \end{aligned}$$

$$2) \int_0^{1/2} \frac{\sqrt{x}}{\sqrt{1-x}} dx = \int_0^{1/2} \frac{x^{\frac{1}{2}} dx}{(1-x)^{\frac{1}{2}}} = \int_0^{\pi/4} \frac{\sin(x)}{(1-\sin^2(x))^{\frac{1}{2}}} \cdot 2 \cos(x) \sin(x) dx$$

$$\begin{aligned}
 x &= \sin^2 \alpha \\
 x' &= 2 \cos(\alpha) \sin(\alpha)
 \end{aligned}$$

$$\frac{dx}{d\alpha} = 2 \cos(\alpha) \sin(\alpha)$$

$$(\Rightarrow) dx = 2 \cos(\alpha) \sin(\alpha) d\alpha$$

$$\begin{aligned}
 0 &= \sin^2 \alpha \\
 (\Rightarrow) \alpha &= 0
 \end{aligned}$$

$$\begin{aligned}
 1/2 &= \sin^2(\alpha) \\
 (\Rightarrow) \frac{\sqrt{2}}{2} &= \sin(\alpha)
 \end{aligned}$$

$$\alpha = \frac{\pi}{4}$$

$$= \int_0^{\pi/4} \frac{\sin(\alpha)}{\sqrt{\cos^2(\alpha)}} \cdot 2 \cos(\alpha) \sin(\alpha) d\alpha$$

$$= \int_0^{\pi/4} \frac{\sin(\alpha)}{\cos(\alpha)} \cdot 2 \cos(\alpha) \sin(\alpha) d\alpha$$

$$= 2 \int_0^{\pi/4} \sin^2(\alpha) d\alpha$$

$$= 2 \int_0^{\pi/4} \frac{1 - \cos(2\alpha)}{2} d\alpha$$

$$= \int_0^{\pi/4} (1 - \cos(2\alpha)) d\alpha$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\Leftrightarrow \cos(2x) = 1 - 2\sin^2(x)$$

$$\Leftrightarrow \frac{1 - \cos(2x)}{2} = \sin^2(x)$$

$$= \int_0^{\pi/4} 1 dx - \int_0^{\pi/4} \cos(2x) dx$$

$$= x \Big|_0^{\pi/4} - \frac{1}{2} \int_0^{\pi/4} 2 \cos(2x) dx$$

$$= \left[ x \right]_0^{\pi/4} - \left[ \frac{1}{2} \cdot \sin(2x) \right]_0^{\pi/4}$$

$$= \left( \frac{\pi}{4} + \frac{1}{2} \cdot \sin(\pi/2) \right) - \left( 0 + \frac{1}{2} \cdot \sin 0 \right)$$

$$= \frac{\pi}{4} + \frac{1}{2} \cdot 1 - 0$$

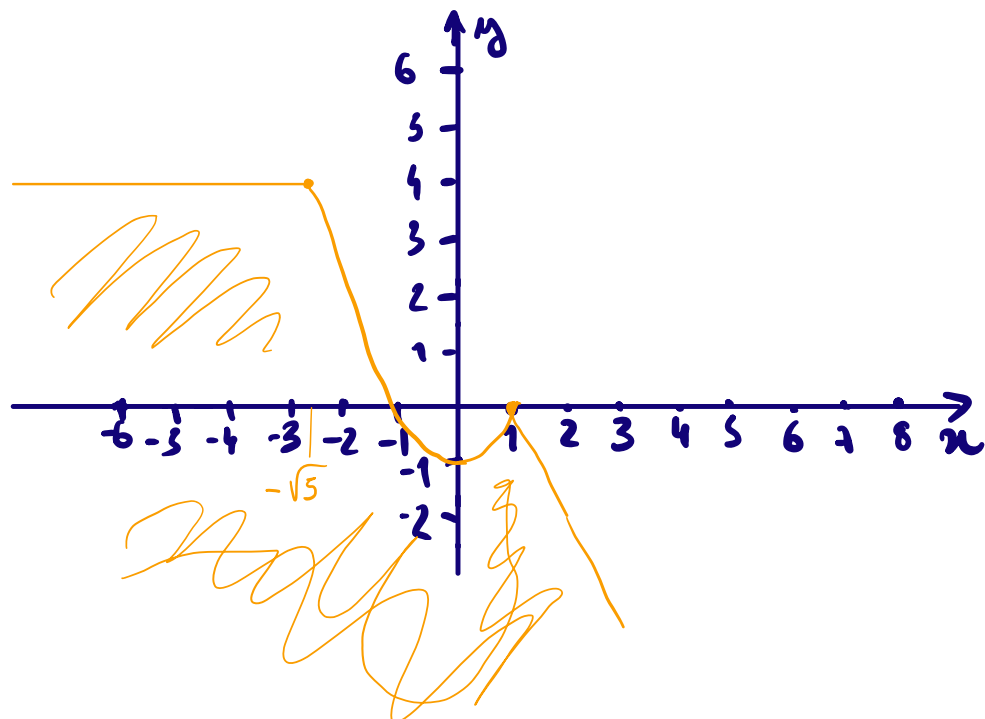
$$= \frac{\pi+2}{4}$$

3) a -

$$y \leq 2 - 2x$$

$$y \leq 4$$

$$y \geq x^2 - 1$$



interseção da reta horizontal com  $y = 2 - 2u$

$$4 = 2 - 2u$$

$$\Rightarrow u = -1$$

$$(-1, 4)$$

interseção da reta horizontal com  $y = u^2 - 1$

$$4 = u^2 - 1$$

$$\Rightarrow 5 = u^2$$

$$\Rightarrow u = \pm\sqrt{5}$$

$$(-\sqrt{5}, 4), (\sqrt{5}, 4)$$

interseção  $y = 2 - 2u$  com  $y = u^2 - 1$

$$2 - 2u = u^2 - 1$$

$$\Rightarrow 0 = u^2 + 2u - 3$$

$$\Rightarrow u = \frac{-2 \pm \sqrt{4 + 12}}{2}$$

$$\Rightarrow u = \frac{-2 \pm 4}{2}$$

$$\Rightarrow u = 1 \quad \vee \quad u = -3$$

$$y = 2 - 2 = 0$$

$$y = 2 + 6 = 8$$

$$(1, 0) \quad (-3, 8)$$

b - De  $-\sqrt{3}$  até  $-1$

$$\int_{-\sqrt{3}}^{-1} (4 - (u^2 - 1)) du = \int_{-\sqrt{3}}^{-1} (5 - u^2) du$$

De  $-1$  até  $1$

$$\int_{-1}^1 (2 - 2u - (u^2 - 1)) du = \int_{-1}^1 (-u^2 - 2u + 3) du$$

$$A = \int_{-\sqrt{3}}^{-1} (5 - u^2) du + \int_{-1}^1 (-u^2 - 2u + 3) du$$

II

1) Calcular a área usando o gráfico

1º triângulo está na parte negativa

$$\text{logo } - \frac{|x|}{2} = -\frac{1}{2}$$

2º triângulo está na parte positiva

$$\text{logo } \frac{|x|}{2} = \frac{1}{2}$$

Quadrado

$1 \times 1$

$$A_{\text{rea}} = -\frac{1}{2} + \frac{1}{2} + 1 = 1$$

b- Pela definição de integral, se o limite inferior e superior forem iguais, o valor é igual a 0

Logo  $n = -1$  é zero

$$\int_{-1}^1 f(u) du = \int_{-1}^0 f(u) du + \int_0^1 f(u) du$$

$$= -\frac{1}{2} + \frac{1}{2} = 0 \quad \text{logo } n \neq 1 \text{ é zero}$$

c-

|        |    |  |   |   |  |   |  |   |  |   |  |   |
|--------|----|--|---|---|--|---|--|---|--|---|--|---|
|        | -1 |  |   | 0 |  | 1 |  | 2 |  | 3 |  | 4 |
| $f(u)$ | -  |  | - | 0 |  | + |  | + |  | - |  | - |
| $F(u)$ | ↘  |  | ↘ | ↗ |  | ↗ |  | ↘ |  | ↘ |  | ↘ |

Logo o mínimo ocorre em  $u = 0$

$$\int_{-1}^0 f(u) du = -\frac{1}{2}$$

O máximo ocorre em  $u = 2$

$$\int_{-1}^2 = 1$$

$$\min F(u) = -1/2$$

$$\max F(u) = 1$$

d- Para ser primitiva a função tem de ser contínua

é necessário definir a função no intervalo  $]2, 3[$

$$m = \frac{-2-1}{3-2} = -3$$

$$\text{logo } y = -3x + b$$

$$\Leftrightarrow 1 = -6 + b$$

$$\Leftrightarrow 7 = b$$

O polinômio tem de ser

$$y = -3x + 7, \quad 2 < x < 3$$

$$e - \int_{-1}^4 g(x) dx = \int_{-1}^2 f(x) dx + \int_2^3 g(x) dx + \int_3^4 f(x) dx$$

$$= 1 + \int_2^3 g(x) dx - 2$$

$$\Leftrightarrow 0 = -1 + \int_2^3 g(x) dx$$

$$\Leftrightarrow 1 = \int_2^3 g(x) dx$$

A melhor forma de garantir a integral num intervalo de comprimento igual a 1 igual a 1 é definir  $g(x)$  como a constante  $c = 1$

$$g(x) = \begin{cases} f(x) & \text{se } x \in [-1, 2] \cup [3, 4] \\ 1 & \text{se } x \in [2, 3] \end{cases}$$

### III

$$1) F(1) = G(1) \Rightarrow F(-1) = G(-1)$$

$$2) |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$F(x) = \begin{cases} \frac{x^2}{2}, & x \geq 0 \\ -\frac{x^2}{2}, & x < 0 \end{cases}$$



pod ser descrito como

$$\frac{x \cdot |x|}{2}$$

Analisemos para  $x \geq 0$

$$\frac{x \cdot x}{2} = \frac{x^2}{2}$$

Para  $x < 0$

$$\frac{x \cdot -x}{2} = -\frac{x^2}{2}$$

$$4) \int \frac{2x^3 + 2x^2 + 2x + 1}{x^2(x^2 + 1)}$$

$$\frac{2x^3 + 2x^2 + 2x + 1}{x^2(x^2 + 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + 1}$$

$$= \frac{Ax(x^2 + 1) + B(x^2 + 1) + (Cx + D)x^2}{x^2(x^2 + 1)}$$

$$= \frac{Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2}{x^2(x^2 + 1)}$$

$$= \frac{(A+C)u^3 + (B+D)u^2 + Au + B}{u^2(u^2+1)}$$

$$\left\{ \begin{array}{l} A+C=2 \\ B+D=3 \\ A=2 \\ B=1 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{l} C=0 \\ D=2 \\ A=2 \\ B=1 \end{array} \right.$$

$$\frac{2u^3+3u^2+2u+1}{u^2(u^2+1)} = \frac{2}{u} + \frac{1}{u^2} + \frac{2}{u^2+1}$$